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X-ray Structure Analysis of a Membrane Protein Complex

Electron Density Map at 3 Å Resolution and a Model of the Chromophores of the Photosynthetic Reaction Center from *Rhodopseudomonas viridis*

X-ray analysis of three-dimensional crystals of the photosynthetic reaction center from the purple bacterium *Rhodopseudomonas viridis* led to an electron density distribution at 3 Å resolution calculated with phases from multiple isomorphous replacement. The protein subunits of the complex were identified. An atomic model of the prosthetic groups of the reaction center complex (4 bacteriochlorophyll *b*, 2 bacteriopheophytin *b*, 1 non-heme iron, 1 menaquinone, 4 heme groups) was built. The arrangement of the ring systems of the bacteriochlorophyll *b* and bacteriopheophytin *b* molecules shows a local 2-fold rotation symmetry; two bacteriochlorophyll *b* form a closely associated, non-covalently linked dimer ("special pair"). A different local 2-fold symmetry axis is observed for the heme groups of the cytochrome part.

Photosynthetic reaction centers are the protein–pigment complexes from photosynthetic membranes that perform the primary charge separation. The RCs† isolated from purple photosynthetic bacteria are the best-characterized with respect to their chemical nature, their composition and their spectroscopic properties (for recent reviews, see e.g. Feher & Okamura, 1978; Hoff, 1982; Parson, 1982). In general they consist of three protein subunits, which are called H (heavy), M (medium) and L (light). In several cases they contain an additional *c*-type cytochrome. One example is the RC from the bacteriochlorophyll *b*-containing purple photosynthetic bacterium *Rhodopseudomonas viridis* with subunits of apparent molecular weights of 38,000 (cytochrome), 35,000 (H subunit), 28,000 (M subunit) and 24,000 (L subunit). The cytochrome subunit contains four (or five) heme groups (Thornber *et al.*, 1980). Further prosthetic groups are four bacteriochlorophyll *b*, two bacteriopheophytin *b*, one non-heme iron atom and probably two quinones (1 menaquinone, 1 ubiquinone) (Thornber *et al.*, 1980; Shopes & Wraight, 1983). After activation by light or by energy transfer from antenna complexes in the membrane, an electron moves within less than ~20 picoseconds from the primary electron donor, which is thought to be a non-covalently linked BChl-*b* dimer ("special pair") to an intermediate acceptor, a BPh-*b* (Holten *et al.*, 1978). Subsequently, within ~230 picoseconds a primary quinone acceptor is reduced; the electron ends up on a secondary quinone acceptor. The oxidized primary donor is reduced by the cytochrome within ~270 nanoseconds (Holten *et al.*, 1978). A monomeric BChl-*b* was postulated

† Abbreviations used: RC, photosynthetic reaction center; BChl-*b*, bacteriochlorophyll *b*; BPh-*b*, bacteriopheophytin *b*; m.i.r., multiple isomorphous replacement.

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between the primary donor and the intermediate acceptor (Shuvalov & Asadov, 1979). The functions of one BChl-*b* and one BPh-*b* remained unclear. Recently, the crystallization of the RC from *R. viridis* was reported (Michel, 1982). The RC is photochemically active in the crystalline state (Zinth *et al.*, 1983; Gast *et al.*, 1983). We started our X-ray work with two major objectives. Firstly, an atomic model of the RC complex in combination with results from spectroscopic investigations is likely to promote understanding the primary charge separation and electron transport in photosynthesis of purple bacteria; the model may also be of relevance to the photosystem II reaction center of green plants, which is known to resemble the RC of purple bacteria with respect to amino acid sequence of at least one protein component (Williams *et al.*, 1983; Hearst & Sauer, 1984), electron acceptors (Hearst & Sauer, 1984), and sensitivity to several herbicides (Stein *et al.*, 1984). Secondly, such a model will extend our knowledge of the structure of membrane proteins. Apart from this RC complex, the only other membrane protein for which three-dimensional crystals suitable for high-resolution X-ray structure analysis were obtained is matrix porin from *Escherichia coli* (Garavito *et al.*, 1983). In this letter we report the spatial arrangement of the prosthetic groups in the RC as the first result of our structure analysis at 3 Å resolution.

The RCs were isolated and crystallized as described (Michel, 1982); during this work the isolation procedure was slightly modified (Zinth *et al.*, 1983). The space group of the crystals is $P4_32_12$, the unit cell dimensions are $a = b = 223{\cdot}5$ Å, $c = 113{\cdot}6$ Å. X-ray reflection intensities were measured in a dark room kept at -2°C. The X-ray source was a Rigaku Denki rotating-anode generator with graphite monochromator and a 0·3 mm collimator. The focus-to-crystal distance was 300 mm; the crystal-to-film distance was 100 mm. For a complete set of intensity data rotation photographs were taken over a total rotation range of 45° around the c^* axis, and of 15° around the a^* axis. To avoid extensive overlap of reflections on one film, the rotation range per film was 0·5° (native crystals) or 0·6° (heavy-atom derivatives). The effective mosaic spread of the crystals was determined using the program OSC (Rossmann, 1979; Schmid *et al.*, 1981). The values ranged from 0·09° for native crystals to 0·17° for some derivatives. As a consequence of the small rotation range and the large mosaic spread, more than half of the reflections on one film were partially recorded. For example, the raw native data set collected from six crystals consisted of 283,499 significantly measured ($I > 2\sigma(I)$) reflections; of these 176,966 or 62·4% were partially recorded. The crystals are rather insensitive to radiation damage, allowing about 20 photographs to be taken; in addition, the crystals kept their alignment almost perfectly during the whole data collection period. These favorable properties greatly simplified the summing of intensities of partially recorded reflections from successive films.

Data reduction was done in two steps: in step 1, films from one crystal were scaled together using fully recorded reflections. The resulting scaling parameters were then used to sum intensities of partially recorded reflections, and to calculate mean intensities for unique reflections. The quality of data obtained from addition of partially recorded reflections was as good as that from fully recorded reflections. The data from the six native crystals had R_{merge} values between 0·055 and 0·078 ($R_{merge} = \sum (I - \langle I \rangle)/\sum I$). In step 2 mean intensities from all crystals of one kind (native or heavy-atom derivative) were scaled together, and new mean values for unique reflections were calculated. Table 1 gives an overview of the reflection data obtained in this way.

Heavy-atom derivatives for multiple isomorphous replacement were prepared by soaking crystals in a solution of heavy-atom compounds in soak buffer (see the legend to Table 1) for three to five days at 18°C. We found seven promising candidates for heavy-atom derivatives that did not disturb the crystalline order, and which, upon visual comparison of equivalent films, showed perceptible intensity differences to native data. About half a data set was collected from each of these. Difference Patterson maps of four mercury derivatives were interpretable and indicated closely similar major heavy-atom binding sites. Two of the mercury derivatives (MDDB, TAMM; see Table 1) were used to calculate an m.i.r. phase set, which allowed us to calculate and interpret a difference Fourier map of the uranium derivative (URNI; see Table 1). A native electron density map at 6 Å resolution already showed molecular boundaries and features that could be clearly identified as helices.

Data collection for the derivatives MDDB, TAMM and URNI was then continued to obtain complete data sets to 3 Å resolution. The determination and refinement of the complete heavy-atom structure was done in the usual way using a program written by Rossmann, modified by TenEyck and Remington, and by us. The refined heavy-atom parameters are listed in Table 2. The correlation between observed and calculated intensity differences between Friedel pairs showed that the correct space group is $P4_32_12$. The overall figure-of-merit for 50347 phases was $\langle m \rangle = 0{\cdot}58$. An electron density map at 3 Å resolution showed

TABLE 1

Data collection overview for photosynthetic reaction centers

Derivative	No. of crystals	No. of unique reflections	Completeness (%) To 3·0 Å	Completeness (%) 3·06–3·0 Å	Crystal-to-crystal R_{merge} (%)
NATI	6	55,061	88·5	63·0	8·37
MDDB†	7	48,193	81·2	49·8	8·41
TAMM‡	10	53,369	82·2	55·1	8·15
URNI§	6	50,771	82·2	53·1	8·68
DAME‖	2	26,372	44·2	26·7	7·90
PCMS¶	3	22,466	38·8	—	7·50

$R_{merge} = \sum (I - \langle I \rangle)/\sum I$.

† 1 mM-2-methyl-1,4-dichloromercuri-2,3-dihydroxybutane in soak buffer (0·1% N,N-dimethyldodecylamine-N-oxide; 1% heptane-1,2,3-triol (high melting point isomer); 1% triethylammonium phosphate; 2·7 M-ammonium sulfate; 20 mM-sodium phosphate (pH 6·5)).

‡ 1 mM-tetrakis·(acetoxymercuri)-methane in soak buffer.

§ 1 mM-uranyl nitrate in modified soak buffer (soak buffer with 4-morpholinoethane sulfonate replacing the phosphate).

‖ 1 mM-ethyl-di(acetoxymercuri)-chloroacetate in soak buffer.

¶ 1 mM-p-chloromercuribenzene sulfonate in soak buffer.

Table 2

Refined heavy-atom parameters

Site		X	Y	Z	Occupancy	β_1	β_2	β_3	β_4	β_5	β_6
MDDB†	1	0·2272	0·0198	0·0298	54·1	11	−135	−633	258	−582	3493
	2	0·9347	0·1356	0·0596	33·0	340	−26	−132	195	−125	718
	3	0·4546	0·3415	0·0737	44·5	316	199	−548	297	−262	692
	4	0·4719	0·3513	0·0463	30·8	81	−30	−471	260	241	188
	5	0·1028	0·2617	0·0955	39·1	283	140	452	116	136	761
	6	0·1084	0·2993	0·1065	31·1	967	−2341	−310	1535	−374	5506
	7	0·4820	0·3658	0·0405	14·5	838	726	−3184	263	−1875	3955
	8	0·4584	0·3394	0·0432	7·2	1817	416	−6562	36	−2479	5036
	9	0·2264	0·0345	0·0511	5·6	787	−2049	−1314	1536	819	701
TAMM	1	0·2277	0·0206	0·0223	54·7	36	17	−603	258	−620	2933
	2	0·4716	0·3510	0·0477	34·1	91	−73	−455	414	145	144
	3	0·4561	0·3419	0·0728	33·4	318	138	−754	165	−114	462
	4	0·4545	0·3389	0·0542	25·7	1606	298	−5355	465	−3394	8092
	5	0·4577	0·3532	0·0554	8·4	32	−108	86	86	52	364
	6	0·1119	0·2943	0·0969	30·1	493	−619	−281	623	−315	2843
	7	0·0339	0·2074	0·0118	8·9	571	342	1062	713	−51	226
	8	0·5995	0·2134	0·1111	3·6	254	125	753	192	807	1519
URNI	1	0·0483	0·0555	0·0356	82·9	1491	962	−1628	1366	−3567	5575
	2	0·0587	0·0441	0·0409	28·4	74	378	−1015	633	−3221	5072
	3	0·4447	0·4414	0·0438	44·8	33	159	73	165	185	1374
	4	0·4399	0·4661	0·0835	44·7	118	−20	−106	213	−66	1090
	5	0·4467	0·4578	0·0572	48·1	125	116	−304	71	14	1502
	6	0·9799	0·0008	0·1072	20·2	116	37	−70	158	173	1381
	7	0·0781	0·0895	0·0394	19·1	217	−131	−18	173	25	1002
	8	0·4435	0·1827	0·0632	6·8	216	29	268	241	−321	2354
	9	0·4098	0·4266	0·0452	11·2	78	49	−688	1119	5195	12570

URO5‡	1	0·0476	0·0545	0·0359	125·4	1845	383	−558	1406	−2454	3325
	2	0·0590	0·0444	0·0401	37·2	138	197	−759	784	−3113	3902
	3	0·4446	0·4414	0·0441	53·2	94	188	−9	204	266	956
	4	0·4402	0·4660	0·0839	55·0	229	−74	−232	273	121	529
	5	0·4470	0·4579	0·0560	58·5	155	110	−300	149	2	811
	6	0·9806	0·0006	0·1064	25·4	174	61	63	225	125	795
	7	0·0780	0·0894	0·0408	19·9	271	−26	0	196	142	604
	8	0·4435	0·1828	0·0621	7·5	304	−4	−475	310	−57	997
	9	0·4098	0·4277	0·0477	21·1	81	58	29	1871	5636	7203
DAME	1	0·2279	0·0207	0·0200	57·7	145	−1	−409	359	−529	2384
	2	0·9346	0·1350	0·0593	66·0	625	−90	−46	358	−54	1706
	3	0·4534	0·3409	0·0743	41·4	707	376	−904	307	−217	1309
	4	0·4721	0·3514	0·0470	19·7	90	−33	−376	384	243	128
	5	0·1025	0·2612	0·0941	51·9	639	243	920	294	323	1558
	6	0·1135	0·2936	0·1087	37·8	877	−1023	1532	1140	−1298	4503
	7	0·4827	0·3671	0·0327	15·1	329	292	−592	250	−498	1509
PCMS	1	0·2281	0·0197	0·0372	46·3	18	59	−143	268	−81	3038
	2	0·4816	0·3641	0·0284	26·9	502	172	448	461	740	2444
	3	0·9348	0·1347	0·0598	56·4	476	−174	104	308	−119	1585
	4	0·4529	0·3401	0·0738	18·4	271	371	−208	209	56	140
	5	0·1019	0·2612	0·0924	47·8	462	242	440	254	154	1550
	6	0·1121	0·2969	0·1005	37·6	425	−510	419	869	−1102	2504
	7	0·4726	0·3512	0·0462	14·2	46	−250	−347	412	388	207
	8	0·2210	0·0234	0·0456	8·2	304	465	−1034	485	1251	1735
	9	0·9074	0·1351	0·0573	8·3	271	−166	−253	98	−382	2056
	10	0·2163	0·0438	0·0671	7·4	215	−89	−45	557	−278	38
	11	0·4420	0·3372	0·0827	15·9	252	−1	258	176	52	1242

X, Y, Z are the fractional co-ordinates of the site. Occupancies are on an arbitrary scale. β_1 to β_6 are the temperature factors according to the expression $T = \exp(-\beta_1 h^2 - \beta_2 hk - \beta_3 kl - \beta_4 k^2 - \beta_5 hl - \beta_6 l^2)$. β values were multiplied by 10^6 for presentation in the Table.

† See Table 1 legend for abbreviations.

‡ URO5, URNI with slightly different soaking conditions (1 crystal).

shape and packing of the RC molecules; large helical regions could be recognized. The solvent region contained evenly distributed noise. To improve the map further, the solvent area was flattened using a procedure developed by Wang (1983). With this procedure the electron density in the solvent region was set to zero. In addition, the distorted density around the major heavy-atom sites also was set to zero. The modified electron density distribution was Fourier-transformed, and the resulting phases were combined with the m.i.r. phases using the phase combination procedure of Hendrickson & Lattman (1970). Phase combination increased the overall figure-of-merit to $\langle m \rangle = 0{\cdot}76$; the improvement of the figure-of-merit was most significant at 3 Å resolution where $\langle m \rangle$ increased from 0·45 to 0·72. The success of solvent flattening is most probably a consequence of the high solvent content of the RC crystals (more than 70%, v/v). The combined phases were used to calculate a new electron density map at 3 Å resolution, which is the basis for the structural details reported below.

A section of the electron density map is shown in Figure 1. The map is of good quality. The asymmetric unit in the crystal contains one RC complex whose molecular boundaries can be clearly recognized. The complex has an elongated

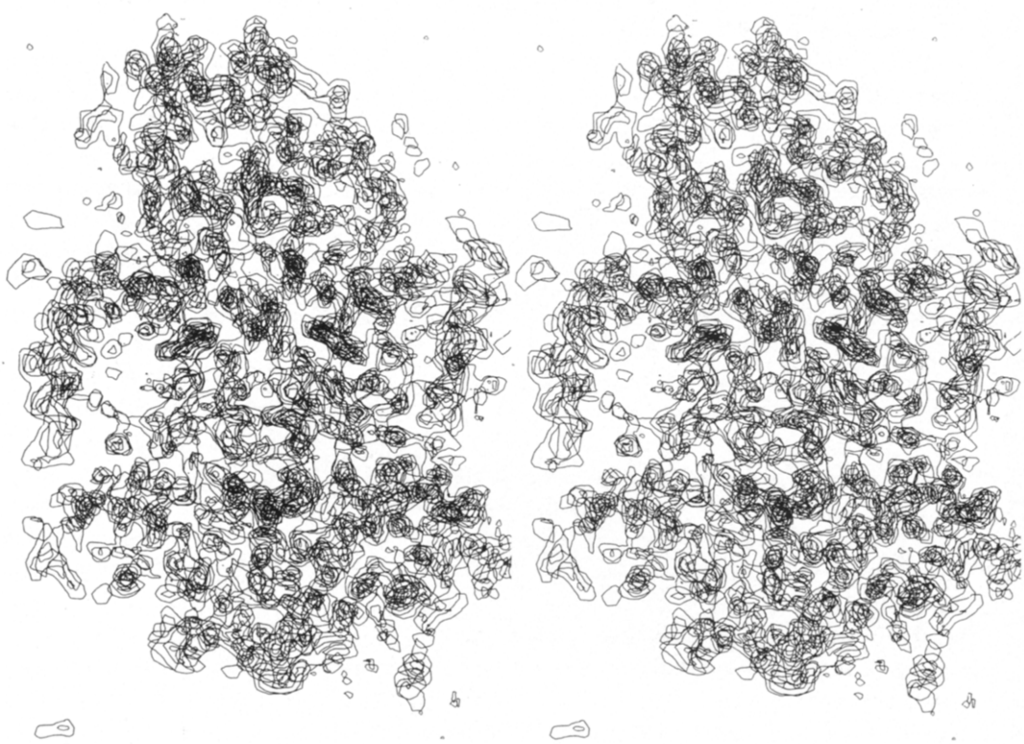


FIG. 1. Stereo drawing (scale 0·09 cm/Å) of 11 layers (distance between layers 1·2 Å) of the electron density distribution at 3 Å resolution. Contour lines start at 1·5 with increments of 1·8 (arbitrary units). The map was rotated so that the central local symmetry axis runs vertically in the middle layer. The section shows mainly part of the central region with some chromophores; small parts of the cytochrome (top), and of the H subunit (bottom) can be seen.

shape within which one can distinguish three regions: the central part has the form of a cylinder with elliptical cross-section (largest diameter ~70 Å, smallest diameter ~30 Å). The length of the cylinder is defined by two flat faces perpendicular to the cylinder axis, which are ~50 Å apart. To each flat face of the cylinder a globular subunit is attached, the centers of these peripheral subunits do not lie on the cylinder axis. Extensions of these attached subunits into the central part can be observed. The largest dimension of the RC complex is the distance of ~140 Å between the outer edges of the peripheral subunits. In the central part we found electron density suitable for four BChl-*b*, two BPh-*b*, one large density peak in which we placed the non-heme iron atom, and density for a quinone molecule. These prosthetic groups are known to be associated with the L and M subunits of the RC of *R. sphaeroides* (Feher & Okamura, 1978), and of *Rhodospirillum rubrum* (Wiemken & Bachofen, 1984). Since there are considerable sequence homologies between the L and M subunits of *R. sphaeroides*, *R. viridis* (Michel *et al.*, 1983) and *R. rubrum* (Theiler *et al.*, 1983), we assume that these subunits account for the protein in the central part of the complex. Within one of the globular subunits extending from the central part we found electron density for four heme groups indicating that this region represents the cytochrome molecule. The globular subunit attached to the opposite side of the central region must therefore be the H subunit of the RC complex.

Model building was done with a Vector General 3400 interactive display system attached to a VAX11/782 computer and the program FRODO (Jones, 1978); the real-space refinement option, which was recently added to FRODO (Jones & Liljas, 1984), was found very useful for optimally fitting the large flat ring systems of BChl-*b*, BPh-*b* and hemes to the density. Experience with other protein structures shows that atomic models built according to an m.i.r. map are usually affected with errors of the order of 0·5 Å to more than 1 Å; distances and angles reported below can therefore be approximate values only. The electron density allows clear distinction between BChl-*b* and BPh-*b* molecules as shown in Figure 2. The BChl-*b* densities are roughly uniform for the complete ring system, whereas the BPh-*b* densities show a deep depression in the middle of the ring system due to the absence of a magnesium ion. Electron density connections from the protein to the centers of the BChl-*b* molecules indicate that one protein side-chain is bound as a ligand to the magnesium ion of each BChl-*b*. Continuous electron density can be found for the complete phytyl chains of three BChl-*b* and one BPh-*b*; parts of the phytyl chains of one BChl-*b* and one BPh-*b* appear to be disordered. The conspicuous electron densities of the phytyl-, and the carbomethoxy groups unequivocally define the orientations of the BChl-*b* and BPh-*b* molecules. The heme groups in the cytochrome part can be easily recognized from their shape and from the high electron density peaks associated with the heme irons in the center (see Fig. 2). Strong density connections from one edge of each heme to the protein indicate the presence of two sulfur bridges per heme as are required for *c*-type cytochromes. They also define the orientations of the heme groups. Each heme iron appears to bind two protein ligands.

The spatial arrangement of the BChl-*b* and BPh-*b* molecules in the central part of the RC complex is characterized by an approximate 2-fold rotation symmetry

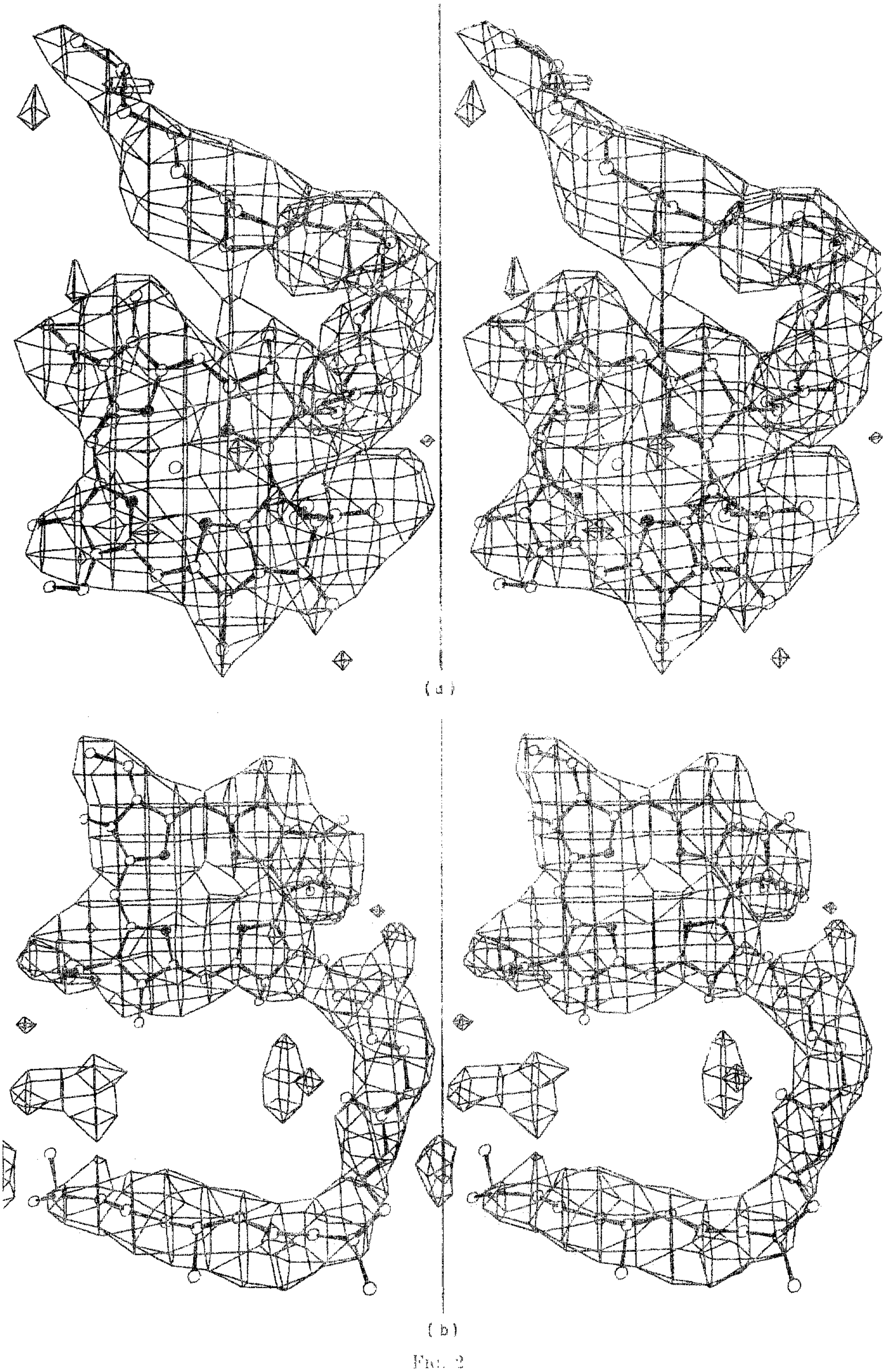
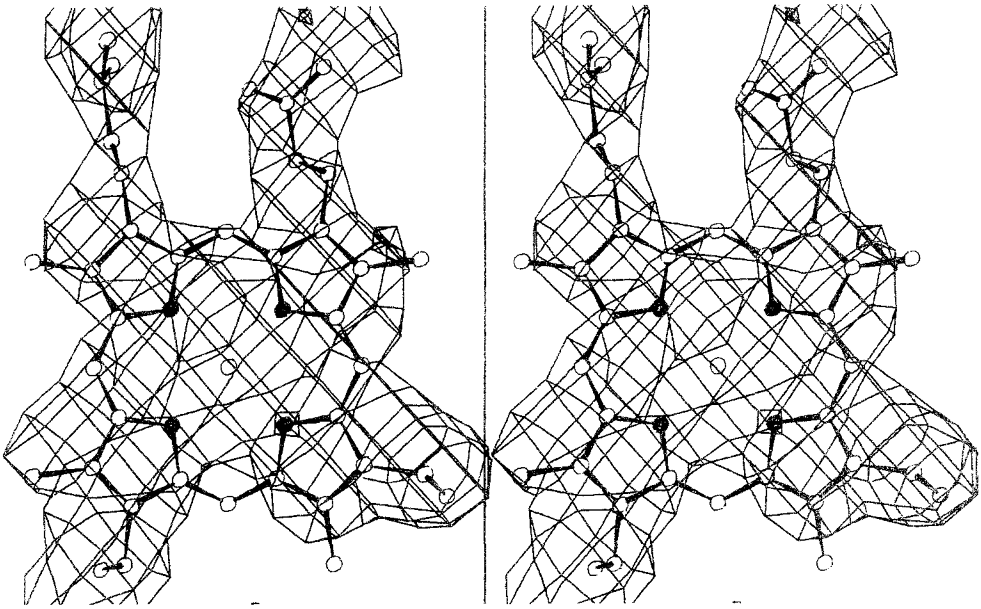


Fig. 2



(c)

FIG. 2. Stereo drawing of chromophore models with electron density (contour level 1·5 arbitrary units): (a) BChl-*b* monomer; (b) BPh-*b*; (c) heme group.

relating two groups, each of which consists of one BChl-*b* of the special pair, one monomeric BChl-*b* and one BPh-*b*. The positions of tetrapyrrole ring atoms and magnesium ions of these two groups can be superposed by the following transformation:

$$\begin{pmatrix} X2 \\ Y2 \\ Z2 \end{pmatrix} = \begin{pmatrix} -0{\cdot}41347 & 0{\cdot}81193 & -0{\cdot}41207 \\ 0{\cdot}76900 & 0{\cdot}06907 & -0{\cdot}63551 \\ -0{\cdot}48753 & -0{\cdot}57965 & -0{\cdot}65293 \end{pmatrix} * \begin{pmatrix} X1 \\ Y1 \\ Z1 \end{pmatrix} + \begin{pmatrix} 121{\cdot}1 \\ -25{\cdot}4 \\ 112{\cdot}6 \end{pmatrix}$$

($X1$, $Y1$, $Z1$) and ($X2$, $Y2$, $Z2$) are co-ordinates of atoms related by the local symmetry, measured in Å in the co-ordinate system defined by the unit-cell axes of the crystal. The root-mean-square deviation of co-ordinates after superposition of 74 atoms is 0·58 Å. In the following we refer to this local rotation axis as the central local symmetry axis. The position of the non-heme iron atom is on or very close to this axis. Two BChl-*b* molecules are closely associated as shown in Figure 3. We interpret this structure as the special pair that was postulated on the basis of results of spectroscopic investigations (Norris *et al.*, 1973,1975; Feher *et al.*, 1973,1975). The two BChl-*b* interact most closely with their pyrrole rings I, which are stacked on top of each other; the average distance between atoms of one of these pyrrole rings to the plane of the other ring is ~3 Å. The acetyl groups of rings I are in direct contact with the Mg atoms of the other BChl-*b* in the pair. Further close contacts exist between the side groups of rings II. The distance between the BChl-*b* Mg ions in the special pair is ~7 Å; the angle between the ring planes is ~15°. The structure of the special pair appears to exclude the possibility

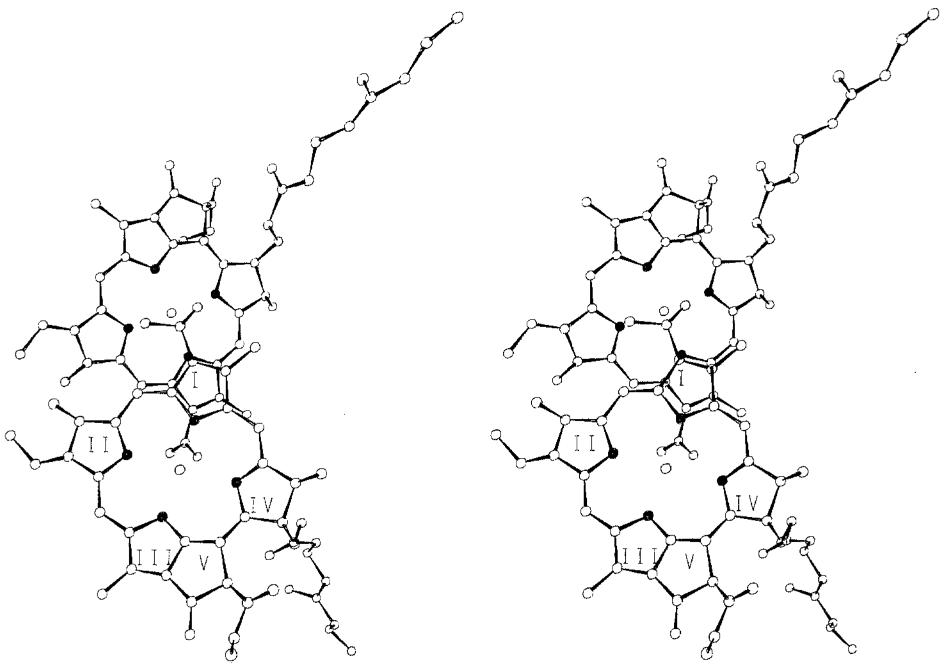


FIG. 3. Stereo drawing of the special pair. The central local symmetry axis runs horizontally between the BChl-*b*s. Ring numbers are indicated in one BChl-*b*. Phytyl chains are truncated.

that a water molecule is involved in the BChl-*b*–BChl-*b* contact in *R. viridis* as was postulated (for a review, see Norris & Katz, 1978). Since chlorophylls *a* and *b* from chloroplasts and cyanobacteria do not possess an acetyl group at pyrrole ring I, the primary donors in photosynthesis of plants and cyanobacteria may be different. Figure 4 shows the arrangement of the BChl-*b*, BPh-*b*, non-heme iron, quinone and cytochrome heme groups in the RC of *R. viridis*. To each BChl-*b* of the special pair another BChl-*b* molecule is associated with Mg–Mg distances of ~13 Å and angles between the tetrapyrrole planes of ~70° (see Fig. 4). Each of these monomeric BChl-*b* molecules is in contact with a BPh-*b*; the distances between the ring centers are ~11 Å, the angles between the ring planes are ~64°. Whereas the tetrapyrrole rings of the monomeric BChl-*b*s and the BPh-*b*s obey the local 2-fold rotation symmetry, parts of their phytyl side-chains do not. Phytyl chains contribute significantly to the contacts between the BChl-*b*s and the BPh-*b*s. A similar observation was made in the structure of a water-soluble light-harvesting BChl-*a* protein from *Prosthecochloris aestuarii* (Matthews *et al.*, 1979).

From the non-heme iron atom four strong electron density connections extend to helical regions of the protein. In three cases the sizes of these connecting densities are consistent with histidine side-chains; the fourth iron ligand appears to be bigger. The only quinone found so far in the electron density also breaks the local 2-fold symmetry of the central axis. Its quinone ring is located near the iron atom (closest distance ~7 Å). The shape of the density is consistent with both

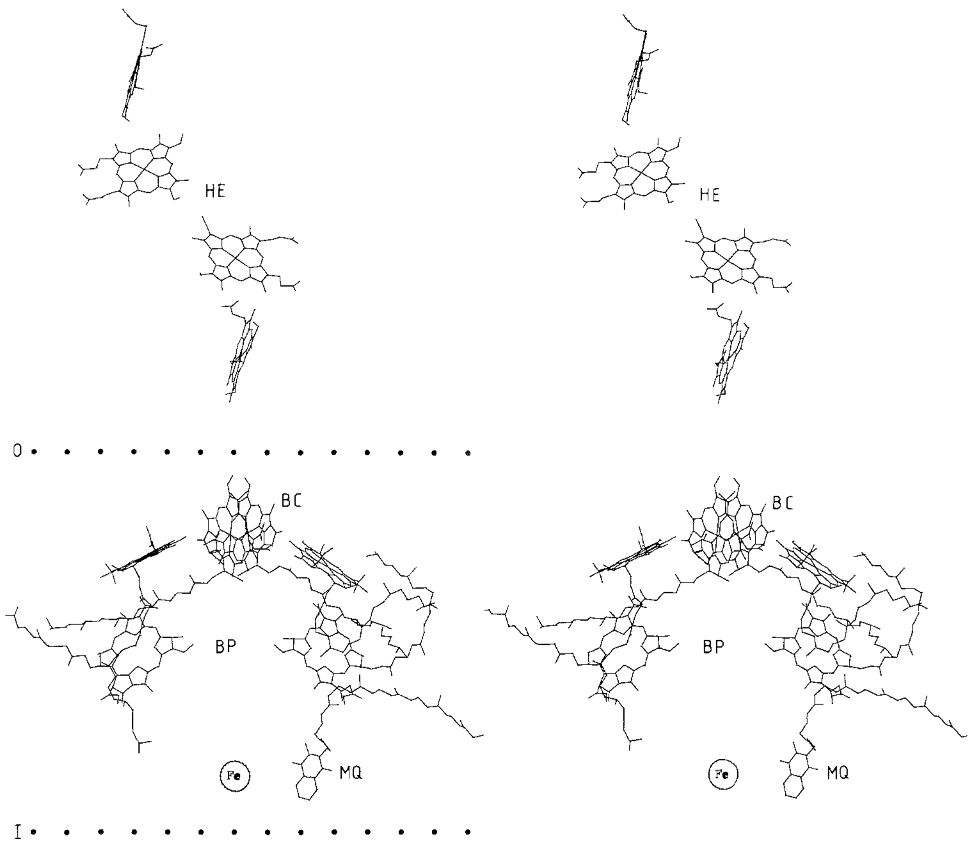


FIG. 4. Stereo drawing of the prosthetic groups of the RC showing 4 BChl-*b* (BC), 2 BPh-*b* (BP), 1 non-heme iron (Fe), 1 quinone (MQ) and 4 heme groups (HE). The central local symmetry axis runs vertically in the plane of the picture. The plane of the membrane is assumed to be oriented perpendicular to the central local symmetry axis, i.e. horizontal and perpendicular to the plane of the picture. The dotted lines marked O and I indicate the presumed approximate outer and inner membrane surfaces of the bacterial cell. The BChl-*b*, BPh-*b*, quinone and iron are located within the cylindrical central part of the RC complex; the ~70 Å diameter of the cylinder is perpendicular to the local symmetry axis in the plane of the picture.

menaquinone and ubiquinone, but from its location we conclude that it is the primary quinone, menaquinone, in *R. viridis* (Shopes & Wraight, 1983). The electron density for the quinone side-chain extends to six of the isoprene units; it approaches one of the BPh-*b*s to van der Waals' distance. The iron atom and the quinone ring are not in direct contact; this observation is consistent with results of recent electron paramagnetic resonance spectroscopic studies (Dismukes *et al.*, 1984). Indirect Fe–quinone interaction *via* one of the Fe ligands appears likely.

The spatial arrangement of BChl-*b*s and BPh-*b*s suggests that an electron can move from the activated special pair to either one of the monomeric BChl-*b*s, and from there to the corresponding BPh-*b*. At the present state of the structure determination both pathways seem to be equivalent. Our present picture is certainly incomplete since it ignores the various interactions of BChl-*b*s and BPh-*b*s with protein residues, which may be different on opposite sides of the central local symmetry axis. Also, at the present state of structure analysis we cannot decide whether deviations of the BChl-*b* and BPh-*b* ring systems from the central local symmetry are real or due to errors in the atomic co-ordinates. The asymmetric location of the quinone speaks strongly against the equivalence of electron pathways. However, we cannot exclude the possibility that one quinone may have been lost during preparation of the crystals, or that quinones are disordered in the crystal. The functional significance of this part of our results remains puzzling. The heme groups in the cytochrome subunit are positioned roughly along a line that is not parallel to the central symmetry axis of the BChl-*b*s and BPh-*b*s. Only the heme closest to the special pair (distances heme–Fe to both BChl-*b*–Mg ~21 Å) lies on this axis. The hemes can be grouped into two pairs, an inner pair that is close to the BChl-*b*s, and an outer pair. These pairs are also related by an approximate 2-fold rotation axis. The direction of this local heme symmetry axis differs by ~60° from that of the central local symmetry axis. The Fe–Fe distances within each heme pair are ~14 Å; the Fe–Fe distance between the neighboring hemes across the local 2-fold axis is ~16 Å. The positions of the heme groups strongly suggest that the oxidized special pair can be reduced only *via* the closest heme.

The detailed interpretation of the protein part of the RC complex has to be postponed until the complete amino acid sequences of its protein subunits are known. The only sequence information presently available is for the 46, 30 and 28 residues at the N termini of the L chain, the M chain and the cytochrome, respectively (Michel *et al.*, 1983). Residues 18 to 46 of the L chain have been located in the central part of the electron density. This supports our assumption that the L subunit forms part of the central region of the RC complex. In this region we can recognize ten helices with a length of ~40 Å each. These helices form two groups of five, which are also related by the central local symmetry axis found for the BChl-*b*s and BPh-*b*s. This observation suggests that the L and M subunits have similar folding, and that each of these subunits contributes five long helices. It should be noted that five highly hydrophobic regions were found in the amino acid sequence of the M subunit of the RC from *R. sphaeroides*, which were interpreted as helical trans-membrane regions (Williams *et al.*, 1983). The directions of all these helices differ from that of the central local symmetry axis by no more than ~30°. An eleventh long helix in the central region of the electron density appears to belong to the H subunit.

From the electron density map we cannot obtain direct evidence for the position and orientation of the membrane relative to the RC complex, because the detergents that replace the membrane are disordered. However, during charge separation in the RC complex an electron is transported across the membrane. Therefore, it is reasonable to assume that the central region of the complex is inserted into the membrane with the special pair close to the outside, and the quinone near the inside of the bacterial cell (see Fig. 4). We believe that the central local symmetry axis runs perpendicular to the plane of the membrane. This axis direction is in agreement with the orientation of the special pair of BChl-*b* molecules found by optical measurements on oriented cells of *R. viridis* (Paillotin *et al.*, 1979).

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